

Radio-Pathomic Maps of Cell Density Identify Brain Tumor Invasion

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Central question. Can routinely acquired multiparametric MRI be linked directly to postmortem histology to estimate tumor cellularity and reveal infiltrative tumor beyond conventional contrast-enhancing margins?

1. Why the study matters

Glioma treatment planning commonly uses post-gadolinium T1 enhancement to define the primary tumor region, yet viable infiltrative tumor can extend well outside enhancement. FLAIR and ADC provide additional clues, but each conventional MRI signature is nonspecific, especially after treatment. The study therefore builds a voxel-level radio-pathomic bridge between MRI and quantitative histology.

2. Cohort, imaging, and pathology

Component	Study design
Patients	44 patients: 32 GBM, 10 non-GBM glioma, 2 other brain tumors
Tissue	93 postmortem tissue samples; 578,668 aligned voxels
MRI	Final clinical T1, T1C, FLAIR, and ADC before death
Pathology	H&E, automated nuclei segmentation and cellularity; selected Ki-67/MIB-1 and CD31 IHC
Model	Bagged random-forest regression with 5 x 5 voxel tiles; 30 model-development and 14 held-out test subjects

3. MRI-histology alignment and modeling workflow

Time between last MR imaging and death (days)	63.0 (62.1)	49.6 (42.3)	111.8 (93.1)	33.0 (13.4)
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Note:—Y/n indicates yes/no.
* Quantitative values are presented as mean (SD).

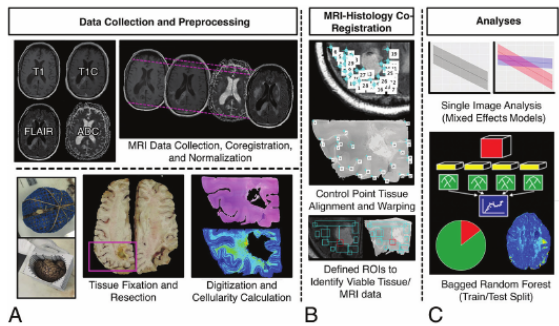


FIG 1. Overview of the data-collection process. A, MR imaging data are collected from the patient's final imaging session before death and coregistered, and T1, T1C, and FLAIR images are in intensity-normalized. Tissue fixation and sampling involve the use of 3D printed brain cages and slicing jigs to preserve structural integrity relative to the MR imaging. Following staining, tissue samples are digitized for cellularity calculation using an automated nuclei segmentation algorithm. B, In-house software is used to align each tissue sample to the FLAIR image using manually defined control points and ROIs. C, Single-image cellularity associations are computed using mixed-effects models, and a bagging regression ensemble is trained to predict cellularity using 1×5 voxel tiles from each image using a two-thirds to one-third train-test split.

orientation information during sectioning. Other issues such as patient's last clinical

Article Figure 1 (p. 3). Clinical MRI, tissue fixation/sampling, digitized histology and cellularity calculation, MRI-histology alignment, mixed-effects analysis and bagged ensemble prediction.

Tissue orientation was preserved with 3D-printed brain cages and slicing jigs. Histology was manually aligned to FLAIR. Single-modality mixed-effects models controlled for time from MRI to death and tissue probability; patient identity accounted for subject-specific effects.

4. Individual MRI sequences

T1, T1C and FLAIR intensity were positively associated with cellularity, while ADC showed the expected negative association. Marginal explained variance was small, however, and associations were generally less pronounced in GBM than in non-GBM glioma - evidence that no single conventional MRI contrast reliably captures tumor cellularity.

5. Radio-pathomic prediction results

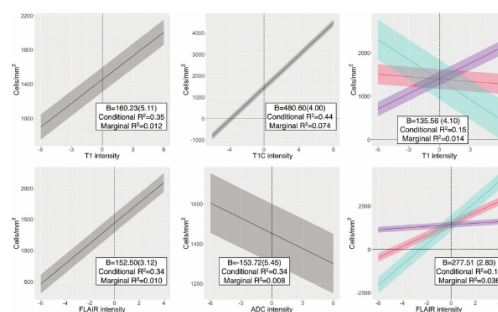


FIG 2. Single-image results depicting the relationship between image intensity and cellularity for each co plots indicate the change in cellularity per SD increase in image intensity and indicate positive associati expected negative association between ADC and cellularity present. β values for the right-handed plots in patients with GBM and NGG and Other patients, indicating that patients with GBM show less pronounced with NGG across all image types, with the exception of T1 intensity.

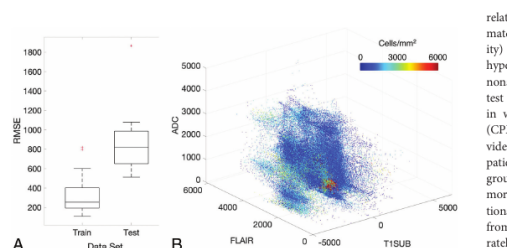


FIG 3. A. Subject-level RMSE values for the training and test data sets. Despite some degree of overfitting, the test set RMSE indicates that the radio-pathomic model is able to accurately predict cellularity across most subjects. B. Sample predictions for test set imaging values presented in terms of their T1SUB, FLAIR, and ADC intensity values. Patterns suggest the presence of traditional imaging signatures but also indicate the lack of specificity for these signatures with regard to hypercellularity. T1SUB indicates TIC-T1.

Article Figures 2-3 (p. 5). Modality-specific cellularity relationships, diagnosis-dependent slopes, subject-level RMSE, and multivariate test predictions in T1SUB, FLAIR and ADC space.

Training RMSE: 389 cells/mm². **Held-out test RMSE:** 1,015 cells/mm². The gap indicates some overfitting and a few test subjects were high-error outliers, but the model generally captured cellularity in unseen subjects.

Its advantage is multivariate integration: several MRI contrasts plus local spatial context are combined rather than treating enhancement, FLAIR, or diffusion as standalone surrogates for tumor.

6. Tumor invasion beyond MRI-defined margins

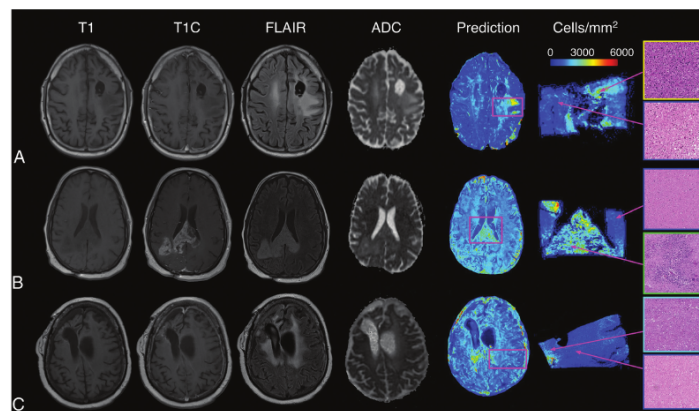
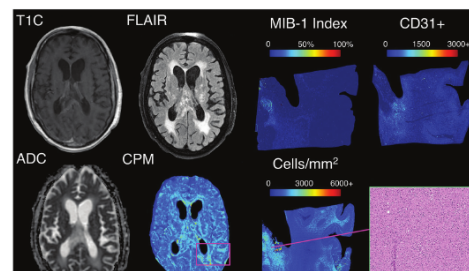


FIG 4. Sample predictions for 3 representative subjects, including a 43-year-old man diagnosed with a grade III anaplastic astrocytoma (A), a 48-year-old man diagnosed with a GBM (B), and a 31-year-old woman diagnosed with a grade III anaplastic astrocytoma (C). These predictions indicate that the radio-pathomic model is able to predict regions of hypercellularity beyond the contrast-enhancing region as well as in the absence of restricted diffusion on the ADC image.



increased angiogenesis throughout the hotspot, confirming likely tumor presence.

DISCUSSION

By means of tissue samples taken at postmortem examination aligned to the patients' clinical imaging, this study assessed relationships between imaging and pathology in the posttreatment state, as well as beyond the contrast-enhancing region. Linear mixed-effects model-based analyses of image intensity values found that single-image signatures explain a relatively small proportion of cellular density.

Article Figures 4-5 (p. 6). Cellularity prediction maps show hypercellular regions outside contrast enhancement and without restricted ADC. Histology/IHC from a nonenhancing hotspot confirms hypercellularity, proliferative activity and angiogenesis.

The most clinically important finding is that cellularity prediction maps can highlight biologically active tumor beyond the conventional enhancing boundary. In the highlighted GBM case, a posterior FLAIR-hyperintense region outside T1 enhancement was predicted to be hypercellular; tissue cellularity, MIB-1/Ki-67 and CD31 supported active proliferating tumor.

7. Interpretation

Enhancement reflects blood-brain-barrier disruption rather than tumor cells themselves, FLAIR mixes edema and infiltration, and low ADC can reflect hypercellularity or necrosis. Pathology-grounded multiparametric modeling therefore offers a richer estimate of invasive disease.

8. Limitations and future directions

The subject-level cohort is small for machine learning despite hundreds of thousands of voxel observations. Cellularity is quantitative but is an indirect and nonexclusive marker of tumor. The interval between final in-vivo MRI and postmortem tissue collection is an important confound because tumor can evolve. The authors suggest that perfusion, spectroscopy and chemical-exchange-saturation-transfer imaging may improve future models.

9. Clinical significance

Because the model uses common T1/T1C/FLAIR/ADC acquisitions, it could be applied retrospectively without additional scan time. With larger validation, radio-pathomic maps could contribute to surgical and radiation targeting, treatment-response monitoring, and spatial studies of resistant or infiltrative tumor.

10. Bottom line

This paper provides pathology-grounded evidence that machine learning can extract information about brain-tumor cellularity that conventional MRI boundaries miss. Its strongest result is the demonstration that **nonenhancing tissue can contain active, hypercellular tumor** and that multiparametric radio-pathomic maps may help localize it.